



Systematic review submitted



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Pandemic literacy and viral zoonotic spillover risk at the frontline of disease emergence in Southeast Asia to improve pandemic preparedness.

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Project Partners

No.	Abbreviations	Full names	Countries	Main roles
1	NMBU	Norwegian University of Life Sciences	Norway	WP7 Lead (Coordinator)
2	NVI	Norwegian Veterinary Institute	Norway	WP4
3	UKHD	University of Heidelberg	Germany	WP5 lead
4	IZW	Leibniz Institute for Zoo and Wildlife Research	Germany	WP3-WP6
5	QMUL	Queen Mary University of London	UK	WP1
6	CU	Chulalongkorn University	Thailand	WP1 lead
7	UMU	Umeå University	Sweden	WP4 lead
8	KKU	Khon Kaen University	Thailand	WP3 lead
9	MU	Mahidol University	Thailand	WP2 lead
10	SUPA71	SUPA71 Co., Ltd.	Thailand	WP6 lead
11	NOVA	NOVA University Lisbon	Portugal	WP9 lead



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2. The material presented and views expressed here are the responsibility of the author(s) only.
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Summary of deliverable 1.3. Systematic review submitted by (UKHD)

UKHD team led by Kate Baernighausen has submitted a manuscript "Public Health Messaging and Community Engagement during COVID-19: A rapid-review from the Greater Mekong Subregion [GMS]" to One Health Outlook on 14 January 2025 UTC with submission ID 59098820-db72-43f4-8e5b-9572577a00de.

In this review, Baernighausen et al., have examined public health messaging strategies and community engagement initiatives implemented in the GMS during the previous pandemic COVID-19.

The Preferred Reporting Items for Systematic Reviews and Meta-Analysis 23 (PRISMA) Checklist was used as methodology to examine Web of Science and PubMed databases.

Inclusion criteria include: COVID-19-related public health messaging; community engagement initiatives, focused on countries within the GMS i.e., Yunnan Province and Guangxi Zhuang Region in China, Myanmar, Thailand, Laos, Vietnam, and Cambodia, published in English between 2020 and 2024, and full-text access is provided. Exclusion criteria include vaccine development and or adverse effects of the pandemic.

An interesting finding is that Vietnam employed comprehensive risk communication strategies and leveraged innovative approaches such as songs and slogans to disseminate public health messages. While the neighbouring Thailand capitulated its pre-existing network of village health volunteers to effectively manage the pandemic's initial phases.

To conclude, although the Greater Mekong Subregion or GMS appears to be homogenous in terms of socioeconomic, cultural and environmental variables, countries and societies vary in their adoption and uptake of public health messaging based on various factors. Thailand and Vietnam, two neighbouring countries employed diverse public health messaging and community engagement strategies during the COVID-19 pandemic. Vietnam mobilised an innovative and inclusive approach to risk communication, while Thailand relied on established volunteer networks, which proved effective initially but fell short in addressing the needs of vulnerable populations, such as migrant workers. This finding emphasizes the importance of adaptable and inclusive strategies to ensure equitable public health outcomes in future pandemics.

Annex 1 includes the approved manuscript.

Annex 1: Copy of the accepted manuscript

1 **Title**

2 Public Health Messaging and Community Engagement during COVID-19: A rapid-review from the
3 Greater Mekong Subregion

4 **Key Words**

5 Rapid review, messaging, community engagement, public health, pandemics

6 **Declarations**

7 Ethical approval: Not applicable.

8 Consent for publication: Not applicable.

9 Availability of data and materials: Not applicable.

10 Competing interests: The authors declare that they have no competing interests.

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14 Abstract

15 Background

16 The COVID-19 pandemic highlighted the importance of public health messaging and community
17 engagement in preventing the spread of disease before vaccines became available. The Greater Mekong
18 Subregion (GMS), a known hotspot for emerging infectious diseases, offers valuable insights that can
19 inform effective public health responses in future pandemics. This rapid review examines public health
20 messaging strategies and community engagement initiatives implemented in the GMS during the
21 COVID-19 pandemic.

22 Methods

23 This rapid review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analysis
24 (PRISMA) Checklist. We used Web of Science and PubMed databases. Articles were included if they
25 addressed COVID-19-related public health messaging and/or community engagement initiatives,
26 focused on countries within the GMS (Yunnan Province and Guangxi Zhuang Region in China,
27 Myanmar, Thailand, Laos, Vietnam, and Cambodia), were published in English between 2020 and
28 2024, and provided full-text access. Articles focusing on unrelated topics, such as vaccine development
29 or adverse effects of the pandemic, were excluded. Data extraction was performed using a calibrated
30 data extraction sheet, with one researcher extracting and another verifying the data.

31 Results

32 After the screening process, 26 articles were included, and 24 were excluded. Most of these articles
33 were published in 2020 learn from measures made in the early phase of the pandemic. Three articles
34 use qualitative methods, five articles use quantitative approaches, eleven articles are identified as
35 descriptive and seven are literature reviews. Vietnam employed comprehensive risk communication
36 strategies, leveraging creative approaches such as songs and slogans to disseminate messages. Thailand
37 utilized its pre-existing network of village health volunteers to effectively manage the pandemic's initial
38 phases.

39 **Conclusion**

40 The GMS employed diverse public health messaging and community engagement strategies during
41 COVID-19. Vietnam demonstrated an innovative and inclusive approach to risk communication, while
42 Thailand relied on established volunteer networks, which proved effective initially but fell short in
43 addressing the needs of vulnerable populations, such as migrant workers. Our findings emphasize the
44 importance of adaptable and inclusive strategies to ensure equitable public health outcomes in future
45 pandemics.